

# Longer Thomason chains via 3-pole composition

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## Abstract

Thomason’s lollipop algorithm turns Smith’s theorem into a procedure: given a Hamiltonian cycle of a cubic graph and one of its edges, it walks through a sequence of Hamiltonian paths and stops at a second Hamiltonian cycle through that edge. The walk can be exponentially long. The best known lower bound, due to Briński and Szady, is  $\Omega(1.1812^n)$  steps on 3-connected planar cubic graphs. We construct 3-connected planar cubic graphs with exactly three Hamiltonian cycles on which the algorithm takes  $\Theta(\rho^{n/6}) = \Theta(1.19742^n)$  steps, where  $\rho = 2.94771\dots$  is the largest root of  $\lambda^4 - 2\lambda^3 - 2\lambda^2 - 2\lambda - 1$ .

The construction comes from a general method for analysing the lollipop walk across 3-edge cuts. We prove that, across a 3-edge cut, the walk either behaves exactly as in the graph with one side contracted, or turns back (the *Mirror Lemma*). Each side of such a cut acts as a finite automaton that does not depend on the rest of the graph (the *Transducer Theorem*). For sides in which every pair of ports is joined by a unique Hamiltonian path, the automata never turn the walk back, and nested copies compose by an exact linear recursion. The growth rate of a nested family is therefore the spectral radius of a small integer matrix computed from a constant-size gadget, here a 7-vertex 3-pole. An exhaustive search shows that no nested family built from a base graph with at most 14 vertices does better.

## 1 Introduction

**Background.** Smith’s theorem, published by Tutte [1], states that every edge of a cubic graph lies in an even number of Hamiltonian cycles. Hence a cubic graph with one Hamiltonian cycle has at least three. Thomason [2] gave a constructive proof, the *lollipop algorithm*. Papadimitriou used it as the motivating example for the complexity class PPA [3]: finding a second Hamiltonian cycle in a cubic graph, given one, is a total search problem in PPA. It is not known to be solvable in polynomial time or to be PPA-complete. Eppstein [10] showed that computing the specific cycle returned by the lollipop algorithm lies in  $\text{FP}^{\text{PSPACE}}$ , and asked whether it is complete for that class.

**Exponential lower bounds.** Since the lollipop algorithm follows a path in an exponentially large implicit graph, it can be slow. Krawczyk [4] gave the first family of planar cubic graphs on which it takes exponentially many steps; Cameron [5] corrected and generalised the construction. Zhong [6] answered a question of Thomassen by giving cyclically 4-edge-connected bipartite families with  $\Theta(2^{n/16})$  steps. Briński and Szady [7] gave 3-connected planar cubic graphs with  $\Omega(1.1812^n)$  steps. This is the best known base [8], and they asked whether families with faster growth exist. On the positive side, the walk takes at most  $n$  steps on cubic bipartite Pfaffian graphs [8]. The fastest known exact algorithm for a second Hamiltonian cycle in a cubic graph runs in  $O(1.2311^n)$  time [9].

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## Our results.

**Theorem 1** (main theorem). *Let  $\rho = 2.9477115868\dots$  be the largest real root of  $\lambda^4 - 2\lambda^3 - 2\lambda^2 - 2\lambda - 1$ . For every  $k \geq 1$  there is a 3-connected planar cubic graph  $G_k$  on  $n = 8 + 6k$  vertices with exactly three Hamiltonian cycles, together with a Hamiltonian cycle  $C_k$  and an edge  $v_0v_1 \in C_k$ , such that Thomason’s algorithm started from  $(C_k, v_0v_1)$  performs exactly  $4 + c_{k-1}$  steps, where  $c_{k-1} = \Theta(\rho^k)$  is given by an explicit linear recursion (Section 7). Consequently it performs  $\Theta(\rho^{n/6}) = \Theta(1.19742^n)$  steps.*

| $n$   | 14 | 20 | 26 | 32  | 38  | 44    | 50    | 56     | 62     | 68      | 74      | 80        |
|-------|----|----|----|-----|-----|-------|-------|--------|--------|---------|---------|-----------|
| steps | 10 | 25 | 73 | 214 | 628 | 1,849 | 5,449 | 16,060 | 47,338 | 139,537 | 411,313 | 1,212,430 |

Table 1: Exact numbers of steps of Thomason’s algorithm on  $G_k$  from  $(C_k, v_0v_1)$ , for  $k = 1, \dots, 12$ . Consecutive ratios approach  $\rho \approx 2.9477$ .

Theorem 1 answers the question of Briński and Szady positively for their class of graphs. More important than the constant, perhaps, is the method behind it. It consists of three structural results about the lollipop walk across a 3-edge cut, which hold for arbitrary cubic graphs:

- **Mirror Lemma** (Theorem 2). Let  $X$  be one side of a 3-edge cut. Then the cycle returned on  $G$  projects either to the cycle returned on  $G/X$ , or to the input cycle; in the second case the two differ only inside  $X$ .
- **Transducer Theorem** (Theorem 4). Every *visit* of the walk to  $X$  has an outcome (the walk passes through or turns back), a successor state and a cost, all determined by  $X$  and the state at entry alone.
- **Simple poles** (Lemmas 5 and 6). If every pair of ports of  $X$  is joined by exactly one Hamiltonian path of  $X$ , the walk never turns back at  $X$ , and the costs of nested 3-poles satisfy an exact linear recursion.

With these, the analysis of a nested family reduces to a finite computation on its base gadget. The certificate is small enough to be checked by hand (Appendix A). The same method makes it easy to search the design space; within it, our family is optimal for base graphs with at most 14 vertices (Section 8).

**Organisation.** Section 2 fixes definitions. Section 3 proves the Mirror Lemma, and Section 4 the Transducer Theorem. Section 5 treats simple poles and composition. Section 6 describes the construction, and Section 7 proves Theorem 1. Section 8 reports the search, Section 9 the computational verification, and Section 10 open problems. Appendix A lists every move of the base gadget, and Appendix B traces a complete example.

## 2 Preliminaries

All graphs are finite and simple. For a vertex  $v$ ,  $N(v)$  is its neighbourhood and  $N[v] = N(v) \cup \{v\}$ . For  $X \subseteq V(G)$ ,  $N(X)$  is the set of vertices outside  $X$  with a neighbour in  $X$ , and  $\delta(X)$  is the set of edges with exactly one end in  $X$ . Paths are written as vertex sequences;  $\text{rev}(S)$  is the reverse of a sequence  $S$ , and  $S \cdot T$  is concatenation. For a path  $P$  and vertices  $u, w$  on it,  $P[u..w]$  is the subpath from  $u$  to  $w$ .

**The lollipop walk.** Let  $G$  be a cubic graph with a Hamiltonian cycle  $C = (v_0, v_1, \dots, v_{n-1})$ , and fix the edge  $v_0v_1$ . A *state* is a Hamiltonian path of  $G$  that starts with  $v_0v_1$ . Let  $P$  be a state with endpoint  $z$ , and let  $zw$  be an edge not on  $P$  with  $w \neq v_0$ . If  $s$  is the successor of  $w$  on  $P$ , the *rotation* of  $P$  along  $zw$  is

$$P' = P[v_0..w] \cdot \text{rev}(P[s..z]),$$

obtained by adding  $zw$  and deleting  $ws$ . Rotations are symmetric:  $P'$  has endpoint  $s$ , and  $P$  is the rotation of  $P'$  along  $sw$ . The *state graph*  $\mathcal{G}(G)$  has the states as vertices and rotations as edges. Every state has degree at most 2, since  $z$  has two non-path edges. A state whose endpoint is adjacent to  $v_0$  has degree at most 1, and adding the edge to  $v_0$  closes it to a Hamiltonian cycle through  $v_0v_1$ . Conversely, each such cycle arises from exactly one such state.

Thomason's algorithm starts at  $S_0 = C - v_{n-1}v_0$ , which is a state of degree 1, and follows its component of  $\mathcal{G}(G)$ . This component is therefore a path, and the algorithm returns the cycle closed by its other end. Operationally, a *step* at endpoint  $z$  uses the unique edge  $zw$  such that  $w$  is neither the predecessor of  $z$  nor the *previous attachment* (the  $w$  of the previous step; initially  $v_0$ ). The algorithm stops when this edge goes to  $v_0$ . We count *steps*, i.e. rotations.

**3-poles and substitution.** A *3-pole* in  $G$  is a set  $X \subset V(G)$  with  $|\delta(X)| = 3$  whose cut edges  $e_i = p_iq_i$  ( $p_i \in X$ ,  $q_i \notin X$ ) form a matching. The  $p_i$  are the *ports*. The graph  $G/X$  obtained by contracting  $X$  to a single vertex  $x$  is then simple and cubic. Let  $h_{ij}(X)$  be the number of Hamiltonian paths of  $G[X]$  between  $p_i$  and  $p_j$ . We call  $X$  *simple* if  $h_{ij}(X) = 1$  for all  $i \neq j$ . For a cubic graph  $F$  and a vertex  $r$ , the graph  $F - r$  is a 3-pole with ports  $N_F(r)$  once it is placed into a host. For a cubic graph  $H$  with a vertex  $y$  and a 3-pole  $P$ ,  $H[y \leftarrow P]$  denotes the graph obtained by deleting  $y$  and joining its three neighbours to the three ports of  $P$  by a fixed bijection, the *wiring*.

**Standing hypothesis.** Whenever we consider a 3-pole  $X$  of a graph in which the lollipop walk runs, we assume  $v_0 \notin X \cup N(X)$ . In particular  $v_1 \notin X$ , and the input cycle crosses  $\delta(X)$  exactly twice.

### 3 The Mirror Lemma

Let  $X$  be a 3-pole of  $G$ ,  $Y = V(G) \setminus X$ , and  $H = G/X$ . A state starts in  $Y$  and alternates between maximal  $Y$ -segments and  $X$ -segments. Since a Hamiltonian path crosses  $\delta(X)$  at most three times, and an even number of times if it ends in  $Y$ , every state has exactly one of the following types. We define a map  $\pi$  from states of  $G$  to states of  $H$  accordingly (Fig. 1).

| type | shape                                                   | cut edges used | $\pi(P)$                |
|------|---------------------------------------------------------|----------------|-------------------------|
| A    | $Y_1 \cdot X \cdot Y_2$ , endpoint in $Y_2$             | 2              | $Y_1 \cdot x \cdot Y_2$ |
| B1   | $Y_1 \cdot X$ , endpoint in $X$                         | 1              | $Y_1 \cdot x$           |
| B3   | $Y_1 \cdot X_1 \cdot Y_2 \cdot X_2$ , endpoint in $X_2$ | 3              | $Y_1 \cdot x \cdot Y_2$ |

Each  $\pi(P)$  is a state of  $H$ , since  $v_0, v_1 \in Y_1$ . For a type B3 state we write  $p_a$  for the entry port of  $X_1$ ,  $p_b$  for its exit port and  $p_c$  for the entry port of  $X_2$ . For a type A state,  $p_{\text{in}}$  and  $p_{\text{out}}$  are the entry and exit ports of the  $X$ -segment.

**Theorem 2** (Mirror Lemma). *Under the standing hypothesis,  $\pi$  maps every edge of  $\mathcal{G}(G)$  either to an edge of  $\mathcal{G}(H)$  or to a single vertex. Consequently, the cycle returned on  $G$  projects either to the cycle returned on  $H$ , or to  $C/X$ . In the second case it agrees with  $C$  outside  $X$ .*

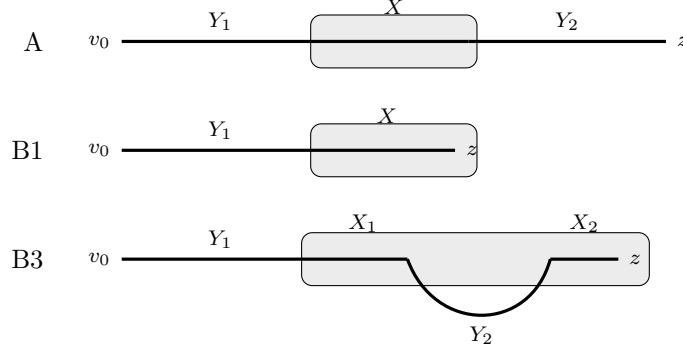


Figure 1: The three types of states relative to a 3-pole  $X$  (shaded). Type B3 uses all three cut edges.

*Proof.* Let  $P \rightarrow P'$  be the rotation at the endpoint  $z$  of  $P$  along  $zw$ , deleting  $ws$ . We use repeatedly that contracting a contiguous block of a sequence commutes with reversing a suffix, provided the cut point  $ws$  is not inside the block.

*Case A* ( $P$  of type A,  $z \in Y_2$ ).

- (i)  $w, s \in Y$ . The  $X$ -segment is untouched or reversed as a whole,  $P'$  has type A, and  $\pi(P')$  is the rotation of  $\pi(P)$  along  $zw$ .
- (ii)  $w \in Y$ ,  $s \in X$ . Then  $ws$  is the entry cut edge, so  $w = q_{\text{in}}$  is the last vertex of  $Y_1$  and  $s = p_{\text{in}}$ . Now  $P' = Y_1 \cdot \text{rev}(Y_2) \cdot \text{rev}(X)$  has type B1, and  $\pi(P') = Y_1 \cdot \text{rev}(Y_2) \cdot x$  is the rotation of  $\pi(P)$  along  $zw$ , which deletes  $wx$ .
- (iii)  $w \in X$ . Then  $zw = e_j$  is the unused cut edge, with  $z = q_j$  and  $w = p_j$ . Since the ports are distinct,  $p_j \neq p_{\text{in}}, p_{\text{out}}$  is an interior vertex of the  $X$ -segment, so  $s \in X$ . Writing the  $X$ -segment as  $X_a \cdot X_b$  with  $X_a$  ending at  $w$ , we get  $P' = Y_1 \cdot X_a \cdot \text{rev}(Y_2) \cdot \text{rev}(X_b)$ , of type B3, with  $\pi(P') = Y_1 \cdot x \cdot \text{rev}(Y_2)$ . In  $H$ , the edge  $q_jx$  is not on  $\pi(P)$ , and the rotation of  $\pi(P)$  along it deletes the edge from  $x$  to the first vertex of  $Y_2$ , giving exactly  $\pi(P')$ .

*Case B1* ( $P = Y_1 \cdot X$ ).

- (i)  $w \in X$ . Then  $s \in X$ , the  $Y$ -part is unchanged, and  $\pi(P') = \pi(P)$ .
- (ii)  $w \in Y$ . Then  $z = p_j$  and  $w = q_j$  for some  $j \neq \text{in}$ . As the  $q_i$  are distinct,  $w$  is not the last vertex of  $Y_1$ , so  $s \in Y_1$ . Writing  $Y_1 = Y_a \cdot Y_b$  with  $Y_a$  ending at  $w$ ,  $P' = Y_a \cdot \text{rev}(X) \cdot \text{rev}(Y_b)$  has type A, and  $\pi(P') = Y_a \cdot x \cdot \text{rev}(Y_b)$  is the rotation of  $\pi(P) = Y_1 \cdot x$  along  $xq_j$ . Here  $q_j \neq v_0$  by the standing hypothesis.

*Case B3* ( $P = Y_1 \cdot X_1 \cdot Y_2 \cdot X_2$ ). All three cut edges are on  $P$ , so  $w \in X$ .

- (i)  $w \in X_2$ . Then  $s \in X_2$  and  $\pi(P') = \pi(P)$ .
- (ii)  $w \in X_1$ ,  $w \neq p_b$ . Then  $s \in X_1$ . Writing  $X_1 = X_{1a} \cdot X_{1b}$  with  $X_{1a}$  ending at  $w$ ,  $P' = Y_1 \cdot (X_{1a} \cdot \text{rev}(X_2)) \cdot \text{rev}(Y_2) \cdot \text{rev}(X_{1b})$  has type B3.
- (iii)  $w = p_b$ . Then  $s = q_b$ , and  $P' = Y_1 \cdot (X_1 \cdot \text{rev}(X_2)) \cdot \text{rev}(Y_2)$  has type A.

In cases B3(ii) and B3(iii),  $\pi(P') = Y_1 \cdot x \cdot \text{rev}(Y_2)$ . Since  $\pi(P) = Y_1 \cdot x \cdot Y_2$  ends at  $q_c$  and  $q_c x$  is not on  $\pi(P)$ , this is the rotation of  $\pi(P)$  along  $q_c x$ , which deletes  $xq_b$ .

No other case occurs, because a non-path edge leads from  $Y$  into  $X$ , or from  $X$  into  $Y$ , only along an unused cut edge. This proves the first claim.

For the consequence, the walk on  $G$  follows the component of  $\mathcal{G}(G)$  from  $S_0(G)$  to its other end  $T$ . Its image under  $\pi$  is a connected subgraph of the component  $L$  of  $\mathcal{G}(H)$  that contains  $\pi(S_0(G)) = S_0(H)$ , and  $L$  is a path whose two ends close to  $C/X$  and to the cycle returned on  $H$ . The endpoint of  $T$  is adjacent to  $v_0$ , so it lies in  $Y$  by the standing hypothesis. Hence  $T$  has type A, and  $\pi(T)$  is a state of  $L$  whose endpoint is adjacent to  $v_0$ , i.e. an end of  $L$ . Finally,  $\pi(T) = S_0(H)$  means that the cycle closed by  $T$  uses the same edges as  $C$  outside  $X$ .  $\square$

## 4 Visits and the Transducer Theorem

**Definition 3** (visits). A *visit* to  $X$  is a maximal run of consecutive steps during which the endpoint lies in  $X$ , together with the step that starts the run (its *entry*) and the step that ends it (its *exit*). By the proof of Theorem 2, a visit starts from a state of type A whose  $X$ -segment is a Hamiltonian path  $Q$  of  $X$ , which we orient from  $p_{\text{in}}$  to  $p_{\text{out}}$ . It has one of two *kinds*:

**B1:** the entry deletes the entry cut edge  $e_{\text{in}}$  (case A(ii));

**B3:** the entry adds the unused cut edge at the third port (case A(iii)).

The *cost* of a B1 visit is the number of steps strictly between its entry and its exit. The cost of a B3 visit is the number of steps after its entry, up to and including its exit. The corresponding passage of the walk on  $H$  through  $x$  takes two steps (B1) or one step (B3), so in both cases the cost is the number of additional steps caused by  $X$ . A visit *transmits* if, after it, the projected walk continues as the walk on  $H$  does; otherwise it *reflects*.

**Theorem 4** (Transducer Theorem). *For every 3-pole  $X$ , every oriented Hamiltonian path  $Q$  of  $X$  between two ports, and every kind  $\kappa \in \{\text{B1}, \text{B3}\}$ , there are an outcome (transmit or reflect), a successor path  $Q'$  and a cost  $c_X(Q, \kappa)$ , such that every visit of kind  $\kappa$  that starts with  $X$ -segment  $Q$ , in any host graph satisfying the standing hypothesis, has this outcome, leaves  $X$ -segment  $Q'$ , and has this cost. Moreover:*

1. *outside visits, every step of the walk on  $G$  projects to the corresponding step of the walk on  $H$ ;*
2. *after a transmitting visit, the walk on  $G$  is at a lift of the state reached by the walk on  $H$ , with the same previous attachment, where ports are identified with  $x$ ;*
3. *a reflecting visit ends at a lift of the state from which its entry was made.*

*Proof. Independence of the host.* While the endpoint is in  $X$ , a step is determined by the endpoint  $z$ , its neighbours, its predecessor and the previous attachment. In type B1 states every attachment lies in  $X$  and its successor too, so only the  $X$ -segment changes (case B1(i)), until the walk leaves along an unused cut edge (case B1(ii)). In type B3 states the steps are of the forms B3(i), B3(ii), which re-splits the  $X$ -pieces and reverses  $Y_2$  as an opaque block, and the exit B3(iii). Hence the sequence of  $X$ -pieces, with their order and orientation, evolves as a function of  $Q$  and  $\kappa$  alone. The previous attachment lies in  $X$  after the first internal step. At the first step of a B1 visit it is  $q_{\text{in}}$ , so that visit cannot leave through  $p_{\text{in}}$  immediately.

*Item 1.* Outside visits the endpoint  $z$  lies in  $Y$ . The edges at  $z$ , its predecessor and the previous attachment correspond in  $G$  and  $H$  (a port corresponding to  $x$ ). So the admissible edge corresponds as well, and by cases A(i)–(iii) the step projects to the step of the walk on  $H$ .

*Items 2 and 3 for B1.* The entry is the step  $S \rightarrow S^*$  of the walk on  $H$  that makes  $x$  the endpoint by deleting  $q_{\text{in}}x$ , after which the previous attachment is  $q_{\text{in}}$ . The exit rotates along some cut edge  $p_jq_j$  and projects to the rotation of  $S^*$  along  $xq_j$  (case B1(ii)), after which the previous attachment is  $q_j$  in both graphs. The walk on  $H$  continues from  $S^*$  along  $xq_c$ , where  $p_c$  is the third port. So an exit through  $p_c$  is that step (transmit), and an exit through  $p_{\text{in}}$  is the rotation along  $xq_{\text{in}}$ , which undoes the entry and returns to  $S$  (reflect).

*Items 2 and 3 for B3.* The entry is the step  $S \rightarrow S^*$  of the walk on  $H$  that attaches  $q_j$  to  $x$ , after which the previous attachment is  $x$ . Steps of the form B3(ii) project alternately to  $S$  and  $S^*$ . The exit B3(iii) lands at a lift of  $Y_1 \cdot x \cdot \text{rev}(Y_2)$ , which is  $S$  or  $S^*$ , with previous attachment  $p_b$ , corresponding to  $x$ . At  $S^*$  the walk on  $H$  continues from there (transmit). At  $S$  the endpoint  $q_j$  has two non-path edges,  $q_jx$  (now forbidden) and one other, so the projected walk goes back along  $L$  (reflect).

The cost is the number of steps of the visit, which by the first part depends only on  $Q$  and  $\kappa$ .  $\square$

## 5 Simple poles

**Lemma 5** (simple poles transmit). *If  $X$  is simple, then every visit to  $X$  transmits.*

*Proof.* A state of  $H$  of type A (endpoint different from  $x$ , with  $x$  interior) has exactly one lift to  $G$ . Its  $Y$ -part is determined, and its  $X$ -segment is the unique Hamiltonian path of  $X$  between the two ports it uses. By Theorem 4(3), a reflecting visit ends at a lift of the state from which its entry was made. That state has type A, so its lift is the state the walk occupied before the visit. The walk on  $G$  would then revisit a state, which is impossible, since it follows a path in  $\mathcal{G}(G)$ .  $\square$

**Lemma 6** (composition). *Let  $P$  be a 3-pole, and let  $X \subset P$  be a simple 3-pole with  $N(X) \subseteq P$ . Let  $P' = P/X$ , with  $X$  contracted to a vertex  $x$ . Then:*

- (a)  $h_{ij}(P) = h_{ij}(P')$  for every pair of ports;
- (b)  $P$  and  $P'$  have the same outcomes and successors, under the bijection of (a);
- (c) for every oriented path  $Q$  of  $P$  and kind  $\kappa$ ,

$$c_P(Q, \kappa) = c_{P'}(Q/X, \kappa) + \sum_f M_{(Q/X, \kappa), f} c_X(f),$$

where  $M_{(Q/X, \kappa), f}$  counts the passages of kind  $f$  through  $x$  during the visit  $(Q', \kappa)$  to  $P'$ . A passage is a step that makes  $x$  the endpoint by deleting  $x$ 's entry edge (kind B1), or a step that attaches to  $x$  (kind B3), together with the oriented pair of ports of  $X$  that the path uses at that moment.

*Proof.* (a) A Hamiltonian path of  $P$  between two ports crosses  $\delta(X)$  exactly twice. So it consists of a Hamiltonian path of  $P'$  together with a Hamiltonian path of  $X$  between the corresponding ports, and the latter is unique.

(b), (c) Fix a host  $G = H_0[y \leftarrow P]$  satisfying the standing hypothesis. Since  $v_0 \notin P \cup N(P)$  and  $N(X) \subseteq P$ , the standing hypothesis also holds for  $X$ , and  $G/X = H_0[y \leftarrow P']$ . By Lemma 5

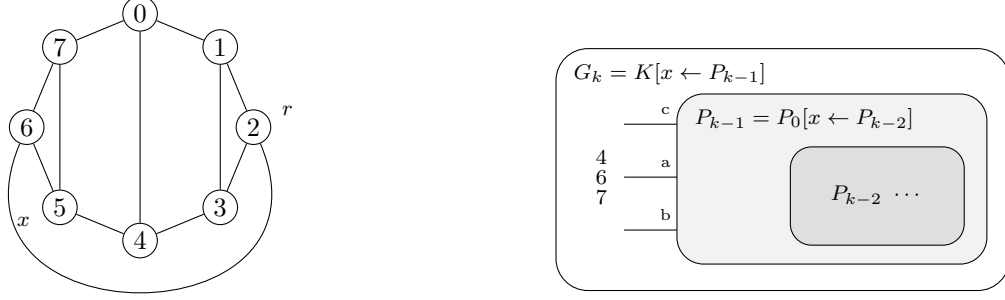


Figure 2: Left: the base graph  $K$ , with chords  $\{0, 4\}, \{1, 3\}, \{5, 7\}$  inside and  $\{2, 6\}$  outside;  $x = 5$  is substituted and  $r = 2$  deleted. Right: the nesting  $G_k = K[x \leftarrow P_{k-1}]$ , where the neighbours  $4, 6, 7$  of  $x$  are joined to the ports  $c, a, b$ .

and Theorem 4, the walk on  $G$  is the walk on  $G/X$  in which each passage through  $x$  is replaced by a transmitting visit to  $X$ . That visit starts in the  $X$ -segment forced by (a), and contributes  $c_X(f)$  extra steps.

Every passage lies inside a visit to  $P'$ , possibly sharing its entry or exit. A B3 passage starts and ends with the endpoint in  $N(x) \subseteq P'$ . A B1 passage either lies inside a visit, or its step is the entry of the visit (when that entry deletes the edge from a port of  $P'$  to  $x$ ); its final step, the rotation at  $x$ , may be the exit of the visit. In every case the extra steps lie strictly inside the window of the corresponding visit to  $P$ . Therefore visits to  $P$  in  $G$  and visits to  $P'$  in  $G/X$  correspond one-to-one, with the same kind, starting state up to contraction, outcome and successor, and their costs differ by the inserted costs. The passages during a visit to  $P'$  are determined by the visit (Theorem 4), so  $M$  does not depend on the host.  $\square$

**Remark 7.** Lemma 6 speaks about visit kinds that occur in some walk. In Section 7 we use only kinds that are reachable from a kind occurring in the walk under consideration, so this is no restriction.

## 6 The construction

**The base graph.** Let  $K$  be the cubic graph on  $\{0, \dots, 7\}$  consisting of the cycle  $012 \dots 70$  and the chords  $\{0, 4\}, \{1, 3\}, \{2, 6\}, \{5, 7\}$  (Fig. 2). It is 3-connected and planar, and has exactly three Hamiltonian cycles. By exhaustive search, it is the only one of the five connected cubic graphs on 8 vertices with exactly three Hamiltonian cycles. Let  $x = 5$  and  $r = 2$ ; then  $r \notin N_K[x] = \{4, 5, 6, 7\}$ . We call the neighbours of  $r$  the ports  $a = 1, b = 3, c = 6$ , and fix the *wiring* that joins the neighbours  $4, 6, 7$  of  $x$  to the ports  $c, a, b$  of the inner pole, respectively.

**The family.** Let  $G_0 = K$  and  $P_0 = K - r$ . For  $k \geq 1$  let  $G_k = K[x \leftarrow P_{k-1}]$  with the wiring above, and  $P_k = G_k - r$ , where  $r$  always denotes the copy of 2 in the outermost copy of  $K$ . Then  $|V(G_k)| = 8 + 6k$ . Since  $r \notin N_K[x]$ ,

$$P_k = (K - r)[x \leftarrow P_{k-1}] = P_0[x \leftarrow P_{k-1}] \quad \text{and} \quad P_k/P_{k-1} = P_0.$$

Let  $C_k$  be the Hamiltonian cycle of  $G_k$  obtained from  $01 \dots 70$  by replacing  $x$  with the unique Hamiltonian path of  $P_{k-1}$  between the ports wired to 4 and 6. The walk starts at  $v_0 = 0$  with the fixed edge  $\{0, 1\}$  of the outermost copy.

| visit | cost | passages through $x$              |
|-------|------|-----------------------------------|
| ab/B1 | 3    | ca/B3                             |
| ab/B3 | 10   | ab/B1, ac/B1, ac/B3, bc/B1, bc/B3 |
| ac/B1 | 5    | ab/B3, ac/B1                      |
| ac/B3 | 10   | ab/B3, ac/B1, ba/B3, bc/B1, cb/B1 |
| ba/B1 | 3    | ca/B1                             |
| ba/B3 | 10   | ab/B3, ac/B1, ca/B3, cb/B1        |
| bc/B1 | 5    | ac/B3, bc/B1                      |
| bc/B3 | 10   | ab/B1, ac/B3, bc/B1, cb/B3        |
| ca/B1 | 3    | ba/B1                             |
| ca/B3 | 6    | ab/B3, ac/B1                      |
| cb/B1 | 3    | cb/B3                             |
| cb/B3 | 6    | ac/B3, bc/B1                      |

Table 2: The twelve visit types of  $P_0 = K - r$ . The first two letters give the oriented pair of ports of the visiting path. All visits transmit, and  $P_0$  is simple. Together the rows define the cost vector  $c_0$  and the passage matrix  $M$ .

**Lemma 8.** *For every  $k$ , the graph  $G_k$  is a 3-connected planar cubic graph with exactly three Hamiltonian cycles, and  $P_k$  is simple.*

*Proof. Simplicity and three cycles.*  $P_0$  is simple (Table 2). If  $P_{k-1}$  is simple, Lemma 6(a), applied to  $P_{k-1} \subset P_k$  with  $P_k/P_{k-1} = P_0$ , shows that  $P_k$  is simple. A Hamiltonian cycle of  $G_k = K[x \leftarrow P_{k-1}]$  consists of a Hamiltonian cycle of  $K$ , which uses a pair of edges at  $x$ , together with the Hamiltonian path of  $P_{k-1}$  between the corresponding ports.  $K$  has three Hamiltonian cycles, and by Smith's theorem each edge lies in exactly two of them, so each pair of edges at  $x$  is used by exactly one. Hence  $G_k$  has exactly three Hamiltonian cycles.

*3-connectivity.* For cubic graphs, 3-connectivity is equivalent to 3-edge-connectivity. Let  $G$  and  $K$  be 3-edge-connected cubic graphs,  $P = G - r$ , and suppose  $F$  is an edge cut of  $K' = K[x \leftarrow P]$  with  $|F| \leq 2$  and sides  $A, B$ . If  $V(P)$  lies on one side, contracting it gives an edge cut of size  $|F|$  in  $K$ , a contradiction. Otherwise let  $F_P = F \cap E(P)$ . Placing  $r$  on either side of the induced partition of  $V(G)$  gives edge cuts of  $G$  of sizes  $|F_P|$  plus the number of ports on the other side, and both are at least 3. Adding the two inequalities gives  $2|F_P| + 3 \geq 6$ , so  $|F_P| = 2 = |F|$ . Then no edge at a port is in  $F$ , so each port lies on the same side as its neighbour in  $K$ . Placing  $x$  on either side therefore gives edge cuts of  $K$  whose sizes are the numbers of ports on the two sides, and both would have to be at least 3, which is impossible with three ports.

*Planarity.* Deleting  $r$  from a plane embedding of  $G_{k-1}$  leaves a face whose boundary meets the three ports in the cyclic order of the rotation at  $r$ . Deleting  $x$  from a plane embedding of  $K$  leaves a face meeting  $N_K(x)$  in the cyclic order of the rotation at  $x$ . Any bijection between two 3-element sets either preserves or reverses cyclic order, and a reversal is undone by mirroring the embedding of  $P_{k-1}$ . So  $P_{k-1}$  can be placed inside that face and joined to  $N_K(x)$  without crossings.  $\square$

## 7 Proof of the main theorem

**The base gadget.** The pole  $P_0 = K - r$  has six oriented pairs of ports, and hence twelve visit types. Table 2 lists, for each, its cost and its passages through  $x$ . All twelve transmit. The passages are named by the ports of the inner pole, through the wiring. Appendix A gives every step.

**Lemma 9.** *For all  $k \geq 1$ ,  $P_k$  has the same outcome table as  $P_0$ , and its cost vector satisfies  $c_k = c_0 + M c_{k-1}$ .*

*Proof.* Induction on  $k$ , applying Lemma 6 to  $X = P_{k-1} \subset P = P_k$ . This is allowed since  $P_{k-1}$  is simple (Lemma 8) and  $N(P_{k-1}) = N_K(x) \subseteq P_k$ , and  $P_k/P_{k-1} = P_0$ .  $\square$

With the visit types ordered ab/B1, ab/B3, ac/B1, ac/B3, ba/B1, ba/B3, bc/B1, bc/B3, ca/B1, ca/B3, cb/B1, cb/B3, we have  $c_0 = (3, 10, 5, 10, 3, 10, 5, 10, 3, 6, 3, 6)$  and

$$M = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

A direct computation gives the characteristic polynomial

$$\det(\lambda I - M) = \lambda^2(\lambda - 1)^2(\lambda + 1)^2(\lambda^2 + 1)(\lambda^4 - 2\lambda^3 - 2\lambda^2 - 2\lambda - 1).$$

The quartic factor is irreducible over  $\mathbb{Q}$ , and its largest real root  $\rho = 2.9477115868\dots$  is the spectral radius of  $M$ . All other eigenvalues have modulus at most 1.17.

*Proof of Theorem 1.* By Lemma 8,  $G_k$  is a 3-connected planar cubic graph on  $8 + 6k$  vertices with exactly three Hamiltonian cycles.

*Exact length.* Apply Theorem 4 to the 3-pole  $X = P_{k-1}$  of  $G_k$ , so  $H = G_k/X = K$ . The standing hypothesis holds, since  $0 \notin N_K[x]$ . On  $K$ , the walk from  $C$  with fixed edge  $\{0, 1\}$  takes 4 steps and passes through  $x$  once, in a passage of type ca/B3. By Lemma 5 the corresponding visit transmits, so

$$\text{steps}(G_k) = 4 + c_{k-1}(\text{ca/B3}).$$

*Lower bound.* Let  $R$  be the set of visit types reachable from ca/B3 in the directed graph of  $M$ . It consists of the ten types other than ba/B1 and ca/B1, and every type in  $R$  occurs in some walk, by induction from the passage above. The integer vector  $u$  on  $R$  given in Table 3 satisfies  $M_R u \geq \frac{2947}{1000} u$  componentwise, where  $M_R$  is  $M$  restricted to  $R$ . This is a finite check, and rows of  $M_R$  refer only to columns in  $R$ . Let  $\varepsilon = \min_{i \in R} c_0(i)/u_i > 0$ . Since  $c_k \geq M c_{k-1}$  and  $M \geq 0$ , induction gives  $c_k \geq \varepsilon(2947/1000)^k u$  on  $R$ . Hence  $\text{steps}(G_k) = \Omega(2.947^k)$ .

*Upper bound.*  $c_k = \sum_{j=0}^k M^j c_0$ , and since  $\rho$  is a simple eigenvalue with all others of modulus less than  $\rho$ ,  $c_k = O(\rho^k)$ .

The lower bound can be sharpened to  $\Omega(\rho^k)$ , since  $\rho(M_R) = \rho$  and  $M_R$  has a positive eigenvector on  $R$ ; the rational certificate in Section 9 gives  $M_R u \geq \frac{29477}{10000} u$ . Hence  $\text{steps}(G_k) = \Theta(\rho^k) = \Theta(\rho^{n/6})$ , and  $\rho^{1/6} = 1.197421\dots$   $\square$

| type  | $u$ |
|-------|-----|
| ab/B1 | 77  |
| ab/B3 | 442 |
| ac/B1 | 227 |
| ac/B3 | 442 |
| ba/B3 | 330 |
| bc/B1 | 227 |
| bc/B3 | 330 |
| ca/B3 | 227 |
| cb/B1 | 77  |
| cb/B3 | 227 |

Table 3: An integer vector with  $M_R u \geq 2.947 u$  componentwise on the reachable set  $R$ .

## 8 Searching the design space

The method gives the exact growth rate of any nested family  $G_k = K[x \leftarrow G_{k-1} - r]$  from the base gadget  $K - r$  alone, provided  $K - r$  is simple and  $r \notin N_K[x]$ . Simplicity forces  $K$  to have exactly three Hamiltonian cycles. The growth rate per vertex is  $\rho(M_R)^{1/(|K|-2)}$ , where  $R$  is the set of visit types reachable from the passages of the walk on  $K$  through  $x$ .

We enumerated all cubic Hamiltonian graphs with at most 14 vertices (as a Hamiltonian cycle plus a perfect matching of chords, up to cospectrality) and kept those with exactly three Hamiltonian cycles. There are 1, 3, 7 and 24 of them on 8, 10, 12 and 14 vertices. For each, we considered all choices of  $x$ ,  $r \notin N[x]$ , wiring, Hamiltonian cycle and start. The best per-vertex rates are 1.197423 (8 vertices, our family), 1.184113 (10), 1.191341 (12) and 1.197423 (14). The best 14-vertex family is two levels of ours. Mixing gadgets from level to level, so that  $c_k = c_0^{(k)} + M^{(k)} c_{k-1}$ , gives the growth rate of a periodic word as  $\rho(M^{(1)} \dots M^{(p)})^{1/\sum(|K_i|-2)}$ . A beam search over periodic words of length up to 8 over all 1,392 distinct gadget matrices with  $|K| \leq 12$  found nothing better than 1.197423. Families whose poles are not simple, so that they may reflect the walk, are outside the exact theory. Direct simulation of several thousand of them with  $|K| \in \{10, 12\}$  gave at most about 1.189 per vertex.

## 9 Verification

Every computational ingredient of the proof is re-derived by a short standalone script, `verify.py`, supplied with this paper. It uses the Python standard library only and runs in under a second. It checks:

- $K$  has exactly three Hamiltonian cycles, and  $P_0$  is simple;
- all twelve visits of  $P_0$  transmit, with the costs and passages of Table 2;
- the growth certificate, in exact rational arithmetic;
- end to end, for  $k \leq 10$ : it builds  $G_k$  explicitly, runs Thomason’s algorithm from  $(C_k, \{0, 1\})$ , and compares the number of steps with  $4 + c_{k-1}(\text{ca/B3})$  (Table 1);
- for small  $k$ : cubic, exactly three Hamiltonian cycles (by exhaustive search), and 3-edge-connected.

We also tested the ingredients independently of this script. An edge-set implementation of the algorithm and a brute-force traversal of the full state graph agree with it on all tested graphs. The Mirror Lemma and the Transducer Theorem were checked on thousands of random hosts and 3-poles. Lemma 6 was checked on 71,484 random pairs  $(P', X)$ , and Lemma 5 on 7,233 random simple poles with up to 21 vertices, without exception. When  $X$  is not simple, the conclusions of Lemma 6 fail in most cases, so the hypothesis cannot be dropped. Planarity, 3-connectivity and the number of Hamiltonian cycles were confirmed directly for  $n \leq 80$ ,  $n \leq 68$  and  $n \leq 50$ , respectively.

## 10 Discussion and open problems

The Mirror Lemma and the Transducer Theorem show that a 3-edge cut behaves like a reversible finite-state gadget that passes the walk through or reflects it. The lollipop walk is then a deterministic reversible particle moving through such gadgets, which is the setting of zero-player reversible motion planning, where PSPACE-hardness results are known [11]. Simple poles never reflect, so they cannot carry information by themselves. Poles with several internal paths can reflect, and their state persists between visits. We believe this is a possible route to Eppstein’s question [10] whether computing the output of the lollipop algorithm is  $\text{FP}^{\text{PSPACE}}$ -complete.

- Is there an exact composition theory for non-simple poles, and do such poles give larger bases?
- Does the method extend to 4-edge cuts? This would be needed for cyclically 4-edge-connected graphs, the class of Zhong’s examples and of Thomassen’s reduction.
- Is there an upper bound of the form  $c^n$  with  $c < 2$  on the length of the lollipop walk in cubic graphs?
- Is the lollipop walk polynomial on cyclically 5-edge-connected cubic graphs?

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## A Every step of the twelve visits of $P_0$

Vertices carry their labels in  $K$ . The column  $Q$  lists the internal path of the visit, from the port nearer  $v_0$ . A B1 visit starts with the endpoint at the first vertex of  $Q$ , with  $Q$  reversed. A B3 visit starts after the cut edge at the third port has been added; the internal path is then split after that port. Each step  $zw$  rotates at the endpoint  $z$  along the edge  $zw$ .  $^\dagger$  marks a step that makes  $x = 5$  the endpoint (a B1 passage), and  $^\ddagger$  a step that attaches to  $x$  (a B3 passage). A  $^\dagger$  on the visit name means that its entry is itself a B1 passage. Every visit transmits.

| visit            | $Q$     | steps after the entry                                                                  | cost | passages through $x$                 |
|------------------|---------|----------------------------------------------------------------------------------------|------|--------------------------------------|
| ab/B1            | 1076543 | 13 40 75 $^\dagger$                                                                    | 3    | ca/B3                                |
| ab/B3 $^\dagger$ | 1076543 | 57 65 $^\ddagger$ 40 76 $^\dagger$ 54 31 07 $^\dagger$ 56 75 $^\ddagger$ 43            | 10   | ab/B1, ac/B1, ac/B3,<br>bc/B1, bc/B3 |
| ac/B1            | 1340756 | 10 45 $^\ddagger$ 76 $^\dagger$ 57 04                                                  | 5    | ab/B3, ac/B1                         |
| ac/B3            | 1340756 | 45 $^\ddagger$ 76 $^\dagger$ 57 01 34 $^\dagger$ 56 70 45 $^\ddagger$ 67 $^\dagger$ 56 | 10   | ab/B3, ac/B1, ba/B3,<br>bc/B1, cb/B1 |
| ba/B1            | 3456701 | 31 04 $^\dagger$ 57                                                                    | 3    | ca/B1                                |
| ba/B3            | 3456701 | 75 $^\ddagger$ 67 04 $^\dagger$ 56 70 13 45 $^\ddagger$ 76 $^\dagger$ 57 01            | 10   | ab/B3, ac/B1, ca/B3,<br>cb/B1        |
| bc/B1            | 3104576 | 34 07 $^\dagger$ 56 75 $^\ddagger$ 40                                                  | 5    | ac/B3, bc/B1                         |
| bc/B3            | 3104576 | 07 $^\dagger$ 56 75 $^\ddagger$ 43 10 76 $^\dagger$ 54 07 65 $^\ddagger$ 76            | 10   | ab/B1, ac/B3, bc/B1,<br>cb/B3        |
| ca/B1            | 6570431 | 67 $^\dagger$ 54 01                                                                    | 3    | ba/B1                                |
| ca/B3            | 6570431 | 10 45 $^\ddagger$ 76 $^\dagger$ 57 04 31                                               | 6    | ab/B3, ac/B1                         |
| cb/B1            | 6754013 | 65 $^\ddagger$ 70 43                                                                   | 3    | cb/B3                                |
| cb/B3            | 6754013 | 34 07 $^\dagger$ 56 75 $^\ddagger$ 40 13                                               | 6    | ac/B3, bc/B1                         |

## B A worked example: the walk on $G_1$

$G_1 = K[x \leftarrow P_0]$  has 14 vertices. Outer vertices are written  $0, \dots, 7$  (without 5), and the inner copy of  $K - r$  as  $0', 1', 3', \dots, 7'$ . The wiring joins  $4-6'$ ,  $6-1'$  and  $7-3'$ . The walk starts from  $C_1$  with fixed edge  $\{0, 1\}$ . Steps marked  $\bullet$  have their endpoint inside the inner pole. Step 1 is the entry of a B3 visit of type ca/B3, and steps 2–7 are its cost  $c_0(\text{ca/B3}) = 6$ , ending at step 7. The total is  $4 + 6 = 10$  steps, as predicted.

| step | rotation $(z, w)$ | path after the step                  |
|------|-------------------|--------------------------------------|
| 0    | –                 | 0 1 2 3 4 6' 5' 7' 0' 4' 3' 1' 6 7   |
| 1    | $(7, 3')$         | • 0 1 2 3 4 6' 5' 7' 0' 4' 3' 7 6 1' |
| 2    | $(1', 0')$        | • 0 1 2 3 4 6' 5' 7' 0' 1' 6 7 3' 4' |
| 3    | $(4', 5')$        | • 0 1 2 3 4 6' 5' 4' 3' 7 6 1' 0' 7' |
| 4    | $(7', 6')$        | • 0 1 2 3 4 6' 7' 0' 1' 6 7 3' 4' 5' |
| 5    | $(5', 7')$        | • 0 1 2 3 4 6' 7' 5' 4' 3' 7 6 1' 0' |
| 6    | $(0', 4')$        | • 0 1 2 3 4 6' 7' 5' 4' 0' 1' 6 7 3' |
| 7    | $(3', 1')$        | 0 1 2 3 4 6' 7' 5' 4' 0' 1' 3' 7 6   |
| 8    | $(6, 2)$          | 0 1 2 6 7 3' 1' 0' 4' 5' 7' 6' 4 3   |
| 9    | $(3, 1)$          | 0 1 3 4 6' 7' 5' 4' 0' 1' 3' 7 6 2   |
| 10   | $(2, 3)$          | 0 1 3 2 6 7 3' 1' 0' 4' 5' 7' 6' 4   |